

DOMAIN-ADAPTIVE EMBEDDING FINE-TUNING FOR FINANCIAL RETRIEVAL IN RAG SYSTEMS

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1 PROJECT SUMMARY

Retrieval-Augmented Generation (RAG) systems rely heavily on the quality of the retrieval component. Even when large language models demonstrate strong generative ability, retrieval errors often propagate into the final responses. Inaccurate retrieval may therefore lead to irrelevant, incomplete, or misleading outputs. As a result, retrieval performance frequently becomes a key bottleneck in practical RAG systems.

This project investigates whether domain-adaptive fine-tuning of embedding models can improve retrieval quality in financial applications. Financial text presents unique challenges, including specialized terminology, implicit reasoning patterns, and semantically subtle distinctions. Embedding models trained on general-domain corpora may not fully capture these characteristics, which can degrade retrieval effectiveness in domain-specific settings.

Our approach focuses on lightweight fine-tuning of an open-source embedding model. The model maps queries and documents into dense vector representations used for similarity-based retrieval. The primary objective of this project is to examine how domain adaptation reshapes the embedding space and whether such adaptation produces measurable improvements in retrieval accuracy.

2 RELATED WORK

Dense retrieval methods have revolutionized modern neural information retrieval by representing queries and documents in a shared latent space. **Dense Passage Retrieval (DPR)** (Karpukhin et al., 2020) first demonstrated that dual-encoder architectures trained with contrastive learning could outperform traditional lexical methods like **BM25**. This established the foundation for embedding-based retrieval in RAG systems. Subsequently, **Sentence-BERT (SBERT)** (Reimers & Gurevych, 2019) introduced siamese Transformer structures to generate semantically meaningful sentence embeddings efficiently.

The challenge of domain mismatch remains a significant hurdle for general-purpose models. **BEIR** (Thakur et al., 2021) provided a comprehensive benchmark showing that zero-shot performance of models like SBERT often degrades on specialized datasets such as **FiQA-2018**. Recent work on the **Finance Massive Text Embedding Benchmark (FinMTEB)** (Tang & Yang, 2025) further highlights that performance on general benchmarks correlates poorly with financial tasks, necessitating the development of domain-adapted models like **Fin-E5**.

To improve the discriminative power of embeddings, research has shifted towards sophisticated training strategies. **RocketQA** (Qu et al., 2021) and **ANCE** (Xiong et al., 2020) emphasized the importance of **hard negative mining**, showing that training on samples that are lexically similar but semantically irrelevant significantly boosts retrieval precision. Furthermore, the **BGE (BAAI General Embedding)** (Xiao et al., 2024) series introduced multi-stage pre-training and fine-tuning pipelines that currently represent the state-of-the-art for open-source retrievers.

Our work differs from these prior studies by specifically focusing on the intersection of **parameter-efficient fine-tuning (PEFT)** and domain-specific financial retrieval. While earlier works often required large-scale re-training, we investigate whether lightweight adaptation of SOTA models (e.g., BGE-small) using the FiQA dataset can yield comparable gains for financial RAG applications with significantly lower compute resources.

Baseline Approach: Our primary baseline is the zero-shot performance of the BAAI/bge-small-en model on the FiQA test set without any domain-specific training.

This baseline allows us to quantify the "out-of-the-box" capability of general-purpose embeddings in the financial domain. We will also consider traditional lexical retrieval (BM25) as a secondary baseline to demonstrate the advantages of dense vector representations in capturing semantic nuances that keyword matching might overlook.

3 PROPOSED METHOD

We propose a domain-adaptive embedding fine-tuning framework specifically designed for financial retrieval within RAG systems. Our approach moves beyond zero-shot inference by incorporating a training-based component, as required by the course guidelines. The model is based on a **Sentence-Transformer dual-encoder** architecture initialized from pretrained state-of-the-art models such as SBERT or BGE.

Neural Network Inputs and Outputs: The network accepts natural language queries q and document passages p as input. Both are processed through the Transformer-based encoder to produce dense vector representations in a shared embedding space $\mathbb{R}^d$.

- **Input:** Tokenized text sequences for query q and document passage p .
- **Output:** Normalized dense embeddings $\mathbf{v}_q, \mathbf{v}_p \in \mathbb{R}^d$ (where $d = 384$ for BGE-small).

Retrieval relevance is determined through similarity computations, specifically the cosine similarity $\text{sim}(\mathbf{v}_q, \mathbf{v}_p)$, between the query and document embeddings.

Loss Function: Training employs **Multiple Negatives Ranking Loss**, a contrastive learning objective. Given a mini-batch of n query-passage pairs $\{(q_i, p_i^+)\}_{i=1}^n$, the loss function is defined as:

$$L = -\frac{1}{n} \sum_{i=1}^n \log \frac{\exp(\text{sim}(\mathbf{v}_{q_i}, \mathbf{v}_{p_i^+})/\tau)}{\sum_{j=1}^n \exp(\text{sim}(\mathbf{v}_{q_i}, \mathbf{v}_{p_j^+})/\tau)}$$

where τ is a temperature hyperparameter. This formulation encourages relevant query-document pairs to be closer in the embedding space while implicitly separating irrelevant pairs through in-batch negatives, allowing for efficient training without explicit negative annotations.

Negative Sampling and Strategy: To improve the model's ability to learn discriminative semantic boundaries, We plan to investigate the inclusion of retrieval-based hard negatives if the standard in-batch negative training exhibits slow convergence or insufficient discriminative power. These hard negatives are passages that are lexically similar to the query but semantically irrelevant, forcing the model to focus on domain-specific nuances. This strategy enhances training effectiveness while maintaining computational efficiency.

Hypothesis: We hypothesize that domain-adaptive fine-tuning improves retrieval accuracy by aligning embedding representations with financial-domain semantics. By updating the model parameters rather than using a frozen baseline, the improved embeddings are expected to better capture specialized terminology and reasoning patterns, leading to measurable gains in retrieval metrics such as Recall@K and MRR.

Evaluation Metrics: We will evaluate retrieval performance using standard information retrieval metrics, specifically **nDCG@10** (Normalized Discounted Cumulative Gain) and **Recall@K** (for $K = 5, 10$). These metrics measure the model's ability to rank the ground-truth relevant document within the top results. Additionally, we will conduct a qualitative analysis by comparing the top-ranked passages retrieved by both the baseline and the fine-tuned models to identify specific types of financial queries where domain adaptation provides the most significant gain.

4 DATASETS / ENVIRONMENTS

We use the FiQA-2018 dataset (Maia et al., 2018) from the BEIR benchmark. This dataset provides a realistic financial-domain retrieval setting consisting of a document corpus, natural language queries,

and relevance judgments. The corpus contains approximately 57K financial text passages, while the query set includes 6,648 questions with corresponding relevance labels.

Data Split: To ensure robust evaluation and monitor for overfitting, we partition the dataset into training, validation, and test sets with a ratio of **80%, 10%, and 10%**, respectively. The training set is used for contrastive learning, the validation set for hyperparameter tuning and early stopping, and the final held-out test set for reporting retrieval performance metrics.

For training, queries are paired with relevant document passages. Negative samples are constructed using both in-batch negatives and retrieval-based hard negatives. Documents are segmented into smaller chunks using a fixed-size windowing strategy (e.g., 512 tokens with 10% overlap) to improve retrieval granularity.

The dataset size is sufficient for batched contrastive training. Fine-tuning remains feasible under modest GPU resources, and we plan to utilize the **GT PACE cluster** or Google Colab Pro. Similarity search is implemented using **FAISS** for efficient nearest-neighbor retrieval.

5 POTENTIAL RISKS

One potential risk is limited improvement after fine-tuning. Retrieval gains may depend on the effectiveness of negative sampling strategies. To address this issue, we will experiment with different hard negative construction methods.

Another risk involves overfitting to the training queries. We mitigate this through validation monitoring, early stopping, and evaluation on held-out subsets of the dataset.

Compute constraints may limit model size and training duration. Therefore, we prioritize lightweight embedding models and parameter-efficient fine-tuning strategies to ensure feasibility within available resources.

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